Eoin Calloran

Senior Embedded Software Engineer

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SUMMARY

Firmware engineer with twelve years of experience in medical and industrial devices. I write safety-conscious C, bring up new boards and debug at the register level. Prefer simple designs that can be certified. I enjoy mentoring and have a strong interest in verification and test automation for firmware. I enjoy simple, reviewable designs and careful documentation, including everything an auditor will ask for.

SKILLS

Embedded C, C++, ARM Cortex-M, FreeRTOS, I2C, SPI, UART, CAN bus, JTAG debugging, IEC 62304, Oscilloscope and logic analyser, Bootloaders, Python

EXPERIENCE

Senior Firmware Engineer, Corrib Medical Devices 2016 – Present

- Developed infusion pump firmware in C on Cortex-M4 under IEC 62304 class C processes, and documented every residual risk with the quality team ahead of the notified body audit.
- Wrote a secure bootloader with signed firmware updates over UART.
- Cut standby current from 180 µA to 35 µA through sleep mode tuning.
- Mentor two junior engineers and review all driver code.
- Introduced automated unit tests and static analysis, reducing defects found in system test by 35%.

Embedded Engineer, Shannon Controls 2012 – 2016

- Implemented Modbus and CAN stacks for industrial drives.
- Created hardware-in-the-loop test rigs.
- Developed a bench test framework in Python that cut regression testing from two days to four hours.
- Collaborated with regulatory staff to support product audits.
- Developed Modbus communication modules that are still deployed in customer plants.

Graduate Engineer, Lough Electronics 2010 – 2012

- Wrote driver code for sensors and displays in C.
- Learned schematic reading and debugging with oscilloscopes.
- Supported production testing of a new product line.
- Contributed to a safety review and implemented fixes for findings in the driver code.
- Created test scripts to validate sensor accuracy on the production line.

EDUCATION

- BEng Electronic Engineering, Western Atlantic University (2008 – 2012)

LANGUAGES

English (native), Irish (B2)